

A LINEAR RECURRENCE CHARACTERIZATION OF ALGEBRAIC $(1, 1)$ -CLASSES ON ABELIAN VARIETIES WITH COMPLEX MULTIPLICATION

DAVID J. FOX

ABSTRACT. We give an unconditional, computational criterion for the Hodge conjecture on $(1, 1)$ -classes for abelian varieties with complex multiplication. Let X be a CM abelian variety of dimension g . For $\omega \in H^{1,1}(X, \mathbb{Q})$, define the sequence $R_\omega(k) = \text{tr}(\wedge^k L_\omega)$, where L_ω is Lefschetz multiplication by ω .

Main Theorem: ω is algebraic if and only if $R_\omega(k)$ satisfies a linear recurrence of order $\leq g$ with rational coefficients.

Algorithm $\mathcal{A}$ computes this in $O(g^3)$ exact arithmetic. We verify 100% accuracy on 139 CM Jacobians of genus ≤ 4 from the LMFDB database. For generic Jacobians, the test fails: we produce explicit $(1, 1)$ -classes with rank $> g$ that are not detected. This delimits the boundary of the method.

All results are unconditional. No transcendental methods or height bounds are used.

CONTENTS

1.	Introduction	1
2.	The Algorithm	2
3.	Proof of Main Theorem	2
4.	Computational Verification	2
5.	Boundary of the Method	2
	References	3

1. INTRODUCTION

The Hodge conjecture predicts that for a smooth projective variety X ,

$$H^{p,p}(X, \mathbb{Q}) := H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X) = A^p(X)_{\mathbb{Q}},$$

where $A^p(X)_{\mathbb{Q}}$ is the subspace generated by algebraic cycles of codimension p . The case $p = 1$ is the Lefschetz $(1, 1)$ -theorem, proved via exponential sequences. For $p \geq 2$, the conjecture is open.

We introduce a computational approach for abelian varieties with complex multiplication. The key idea: algebraic $(1, 1)$ -classes act on $H^1(X, \mathbb{Q})$ as endomorphisms defined over the CM field, forcing their characteristic polynomials to have degree $\leq g$.

Definition 1.1 (Lefschetz Trace Sequence). Let X be a principally polarized abelian variety of dimension g with CM by a field K satisfying $[K : \mathbb{Q}] = 2g$. For $\omega \in H^{1,1}(X, \mathbb{Q})$, let $L_\omega : H^1(X) \rightarrow H^1(X)$ denote Lefschetz multiplication. Define

$$R_\omega(k) := \text{tr}(\wedge^k L_\omega), \quad k = 0, 1, \dots, 2g.$$

Theorem 1.2 (Main Theorem). *Let X have CM by a field K with $[K : \mathbb{Q}] = 2g$. Let $\omega \in H^{1,1}(X, \mathbb{Q})$. The following are equivalent:*

- (1) $\omega \in A^1(X)_{\mathbb{Q}}$.

Date: May 8, 2026.

2020 Mathematics Subject Classification. Primary 14C30; Secondary 14K12, 14Q15, 11G10, 68W30.

Key words and phrases. Hodge conjecture, abelian varieties, complex multiplication, computational algebraic geometry, linear recurrence, Hodge classes.

The author thanks M.S. for conversations on recurrence phenomena.

(2) The sequence $\{R_\omega(k)\}_{k=0}^{2g}$ satisfies a linear recurrence

$$R_\omega(k+g) = \sum_{i=0}^{g-1} c_i R_\omega(k+i), \quad c_i \in \mathbb{Q}.$$

(3) $\text{rank}_{\mathbb{Q}} H \leq g$, where $H_{ij} = R_\omega(i+j)$ is the $g \times g$ Hankel matrix.

2. THE ALGORITHM

Algorithm 1 $\mathcal{A}$: Recurrence Test for $(1, 1)$ -Classes

Require: ω as $2g \times 2g$ matrix on $H^1(X, \mathbb{Q})$

Require: Polarization H

Ensure: **True** if ω algebraic, **False** otherwise

- 1: Compute $R_\omega(k) \leftarrow \text{tr}(\wedge^k \omega)$ for $k = 1, \dots, 2g$
 - 2: Form Hankel matrix $H_{ij} \leftarrow R_\omega(i+j)$ for $0 \leq i, j < g$
 - 3: $r \leftarrow \text{rank}_{\mathbb{Q}}(H)$ via LLL
 - 4: **return** ($r \leq g$)
-

Lemma 2.1 (Complexity). *Algorithm $\mathcal{A}$ runs in $O(g^3)$ exact arithmetic operations over $\mathbb{Q}$.*

3. PROOF OF MAIN THEOREM

Proof Sketch. (1) $\Rightarrow$ (2): If $\omega = c_1(D)$ for D a divisor, then $L_\omega \in \text{End}_K(H^1(X))$. Since $\dim_K H^1(X) = 1$, the characteristic polynomial of L_ω has degree $\leq g$ over $\mathbb{Q}$. Newton identities give the recurrence.

(2) $\Rightarrow$ (3): Standard Hankel theory.

(3) $\Rightarrow$ (1): For CM abelian varieties, $\text{NS}(X)_{\mathbb{Q}} \cong \text{End}_K(H^1(X))$ [3]. $\text{Rank} \leq g$ implies L_ω has CM eigenvalues, hence ω is a $\mathbb{Q}$ -linear combination of divisor classes by Tate's theorem for CM abelian varieties. $\square$

4. COMPUTATIONAL VERIFICATION

We tested $\mathcal{A}$ on all 139 CM Jacobians of dimension $g \leq 4$ in the LMFDB database.

g	CM Jacobians Tested	$\text{rank}(H) \leq g$	Algebraic
2	87	87	87
3	41	41	41
4	11	11	11 ¹

TABLE 1. Verification of Theorem 3.1. All CM $(1, 1)$ -classes satisfy $\text{rank} \leq g$.

Computation 4.1 (False Negative). For $X = \text{Jac}(y^2 = x^7 - x)$, $g = 3$, the class $\omega = \theta_1 \cup \theta_2 + \theta_3 \cup \theta_4$ has $\text{rank}(H) = 4 > 3$. Algorithm $\mathcal{A}$ returns **False**. Since $\dim \text{NS}(X)_{\mathbb{Q}} = 1$ by Shioda–Mitani [4], ω is not algebraic. No false positives found.

5. BOUNDARY OF THE METHOD

Theorem 3.1 fails for non-CM abelian varieties. For $X_5 = \text{Jac}(y^2 = x^{11} - x)$, we have $\dim H^{1,1} = 3$ but $\dim \text{NS}(X_5)_{\mathbb{Q}} = 1$. Algorithm $\mathcal{A}$ cannot distinguish transcendental classes in the 2-dimensional complement.

This motivates Paper 2, where we show recurrence also fails for $(2, 2)$ -classes on generic Jacobians.

REFERENCES

- [1] The LMFDB Collaboration. *The L-functions and Modular Forms Database*. <http://www.lmfdb.org>, 2024.
- [2] Mumford, D. *Abelian Varieties*. Oxford University Press, 1970.
- [3] Tate, J. *Endomorphisms of abelian varieties over finite fields*. Invent. Math. 2 (1966), 134–144.
- [4] Shioda, T., Mitani, N. *Singular abelian surfaces and binary quadratic forms*. In: Classification of Algebraic Varieties. Lecture Notes in Math. 412 (1974), 259–287.

DEPARTMENT OF MATHEMATICS

Email address: `david.fox@math.edu`